



COMP47350: Data Analytics (Conv)

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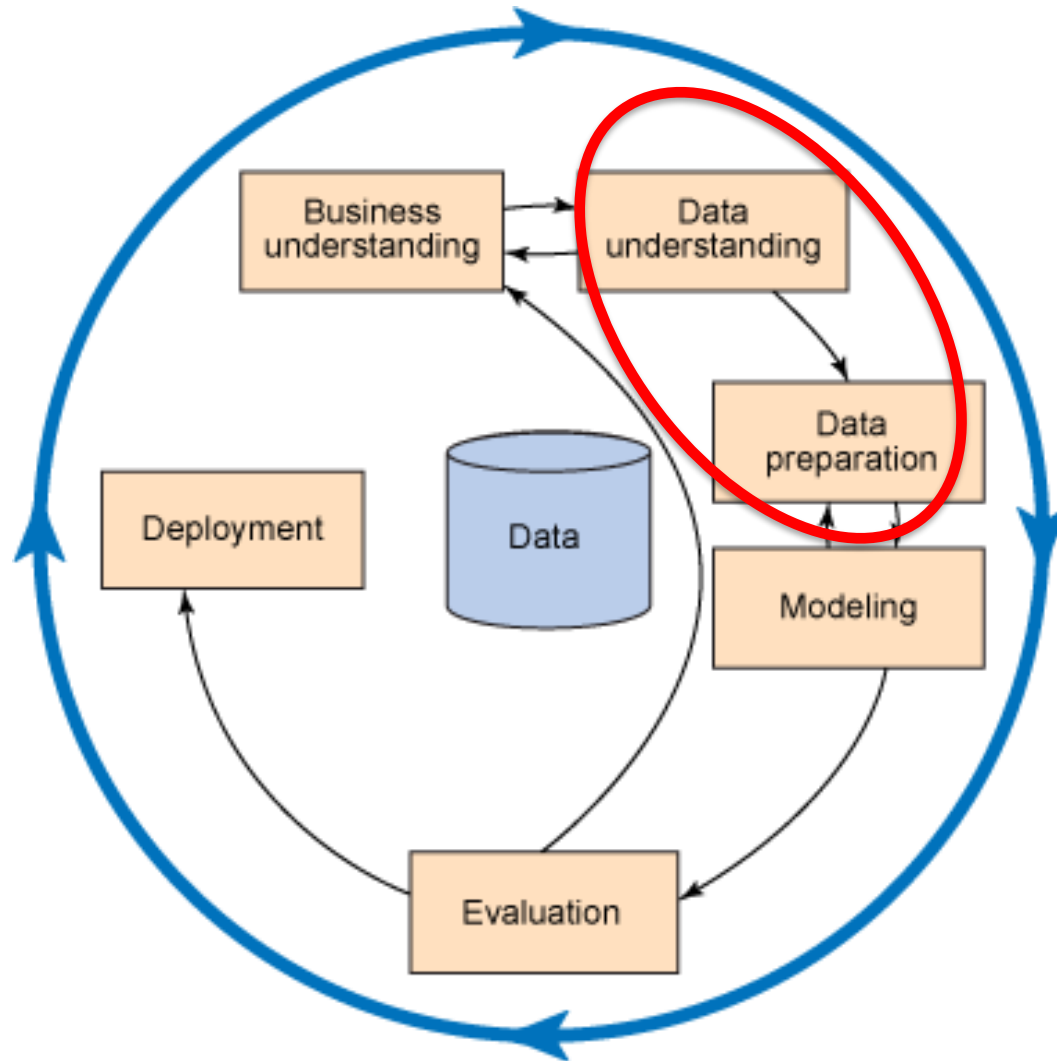
University College Dublin

2025/26

Data Analytics Project Lifecycle:

CRISP-DM

CRISP-DM: **C**Ross-**I**ndustry **S**tandard **P**rocess for **D**ata **M**ining



Data Understanding

1. Data Quality Report (tables, plots)

- The goal of this step is information gathering
- **Short written report** (no code included, to be shared with management)

2. Data Quality Plan

- Handling Data Quality Issues (e.g., missing values, outliers, irregular cardinality)
- The goal of this step is listing actions to implement on each feature
- **Short summary table** listing features and **actions for each feature**

Case Study

Motor Insurance Fraud Prediction

- Examine the **Data Quality Report** for the motor insurance fraud prediction problem and analyze each feature.
- Prepare the **Data Quality Plan** based on observations from the Data Quality Report

Motor Insurance Fraud: ABT Subset

ID	TYPE	INC.	MARITAL STATUS	NUM CLMNTS.	INJURY TYPE	HOSPITAL STAY	CLAIM AMNT.	TOTAL CLAIMED	NUM CLAIMS	NUM SOFT TISS.	% SOFT TISS.	CLAIM AMT RCVD.	FRAUD FLAG
1	CI	0	Married	2	Soft Tissue	No	1,625	3250	2	2	1.0	0	1
2	CI	0		2	Back	Yes	15,028	60,112	1		0	15,028	0
3	CI	54,613		1	Broken Limb	No	-99,999	0	0	0	0	572	0
4	CI	0		4	Broken Limb	Yes	5,097	11,661	1	1	1.0	7,864	0
5	CI	0		4	Soft Tissue	No	8869	0	0	0	0	0	1
6	CI	0	Single	1	Broken Limb	Yes	17,480	0	0	0	0	17,480	0
7	CI	52,567		3	Broken Limb	No	3,017	18,102	2	1	0.5	0	1
8	CI	0		2	Back	Yes	7463	0	0	0	0	7,463	0
9	CI	0	Married	1	Soft Tissue	No	2,067	0	0	0	0	2,067	0
10	CI	42,300		4	Back	No	2,260	0	0	0	0	2,260	0
		:				:					:		
300	CI	0	Married	2	Broken Limb	No	2,244	0	0	0	0	2,244	0
301	CI	0		1	Broken Limb	No	1,627	92,283	3	0	0	1,627	0
302	CI	0		3	Serious	Yes	270,200	0	0	0	0	270,200	0
303	CI	0		1	Soft Tissue	No	7,668	92,806	3	0	0	7,668	0
304	CI	46,365		1	Back	No	3,217	0	0		0	1,653	0
		:				:					:		
458	CI	48,176	Married	3	Soft Tissue	Yes	4,653	8,203	1	0	0	4,653	0
459	CI	0		1	Soft Tissue	Yes	881	51,245	3	0	0	0	1
460	CI	0		3	Back	No	8,688	729,792	56	5	0.08	8,688	0
461	CI	47,371	Divorced	1	Broken Limb	Yes	3,194	11,668	1	0	0	3,194	0
462	CI	0		1	Soft Tissue	No	6,821	0	0	0	0	0	1
		:				:					:		
491	CI	40,204	Single	1	Back	No	75,748	11,116	1	0	0	0	1
492	CI	0		1	Broken Limb	No	6,172	6,041	1		0	6,172	0
493	CI	0		1	Soft Tissue	Yes	2,569	20,055	1	0	0	2,569	0
494	CI	31,951	Married	1	Broken Limb	No	5,227	22,095	1	0	0	5,227	0
495	CI	0		2	Back	No	3,813	9,882	3	0	0	0	1
496	CI	0	Married	1	Soft Tissue	No	2,118	0	0	0	0	0	1
497	CI	29,280		4	Broken Limb	Yes	3,199	0	0	0	0	0	1
498	CI	0		1	Broken Limb	Yes	32,469	0	0	0	0	16,763	0
499	CI	46,683	Married	1	Broken Limb	No	179,448	0	0		0	179,448	0
500	CI	0		1	Broken Limb	No	8,259	0	0	0	0	0	1

Data Quality Report

Motor Insurance Fraud: Tables

Summarise each feature

(a) Continuous Features

Feature	Count	%		Min	1 st		Median	3 rd		Std. Dev.
		Miss.	Card.		Qrt.	Mean		Qrt.	Max	
INCOME	500	0.0	171	0.0	0.0	13,740.0	0.0	33,918.5	71,284.0	20,081.5
NUM CLAIMANTS	500	0.0	4	1.0	1.0	1.9	2	3.0	4.0	1.0
CLAIM AMOUNT	500	0.0	493	-99,999	3,322.3	16,373.2	5,663.0	12,245.5	270,200.0	29,426.3
TOTAL CLAIMED	500	0.0	235	0.0	0.0	9,597.2	0.0	11,282.8	729,792.0	35,655.7
NUM CLAIMS	500	0.0	7	0.0	0.0	0.8	0.0	1.0	56.0	2.7
NUM SOFT TISSUE	500	2.0	6	0.0	0.0	0.2	0.0	0.0	5.0	0.6
% SOFT TISSUE	500	0.0	9	0.0	0.0	0.2	0.0	0.0	2.0	0.4
AMOUNT RECEIVED	500	0.0	329	0.0	0.0	13,051.9	3,253.5	8,191.8	295,303.0	30,547.2
FRAUD FLAG	500	0.0	2	0.0	0.0	0.3	0.0	1.0	1.0	0.5

Data Quality Report

Motor Insurance Fraud: Tables

(a) Categorical Features

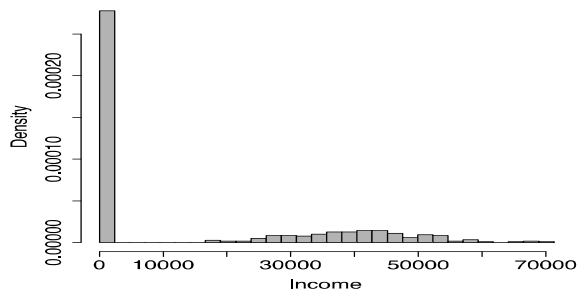
Feature	Count	%		Mode	Mode	Mode	2 nd	2 nd	2 nd
		Miss.	Card.		Freq.	%	Mode	Mode	Mode
INSURANCE TYPE	500	0.0	1	CI	500	1.00	–	–	–
MARITAL STATUS	500	61.2	4	Married	99	51.0	Single	48	24.7
INJURY TYPE	500	0.0	4	Broken Limb	177	35.4	Soft Tissue	172	34.4
HOSPITAL STAY	500	0.0	2	No	354	70.8	Yes	146	29.2

Data Quality Report

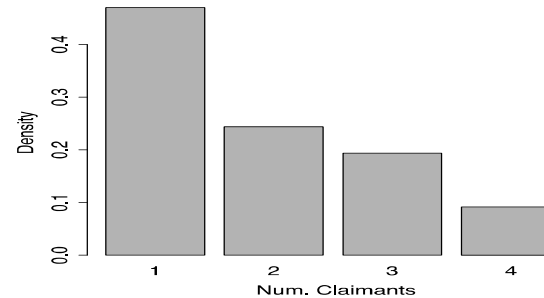
Motor Insurance Fraud: Visualisations (Plots)

For each continuous feature: histogram, boxplot

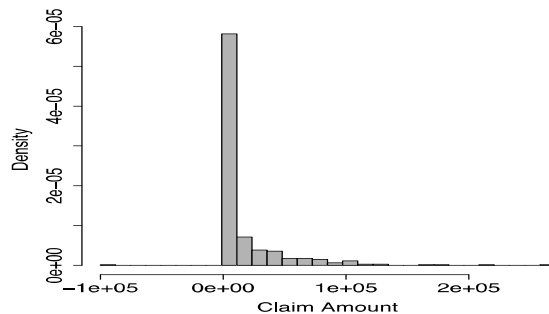
For each categorical feature: barplot



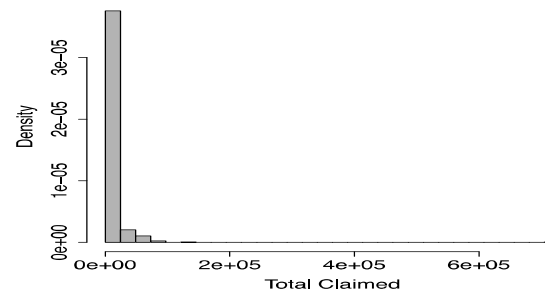
(a) INCOME



(b) NUM CLAIMANTS

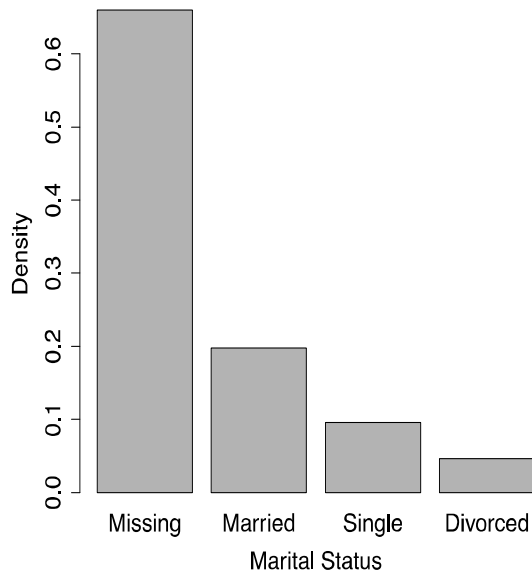


(c) CLAIM AMOUNT

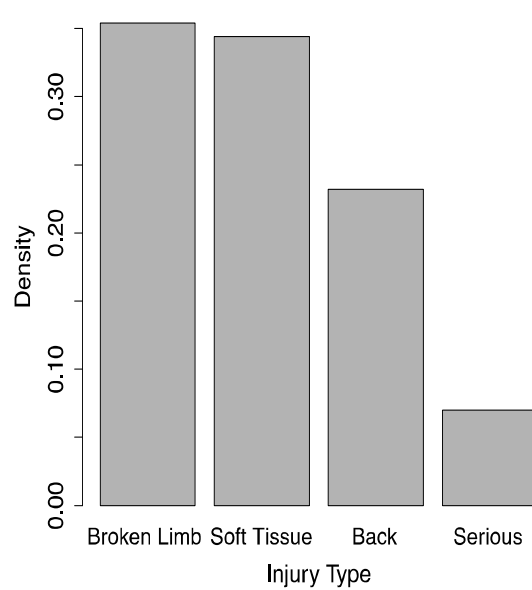


(d) TOTAL CLAIMED

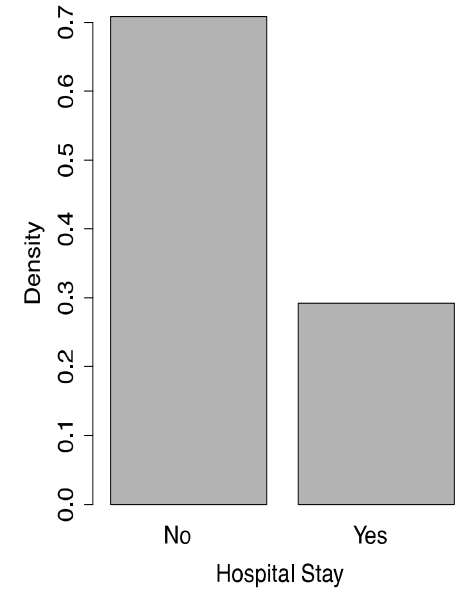
Data Quality Report



(a) MARITAL STATUS

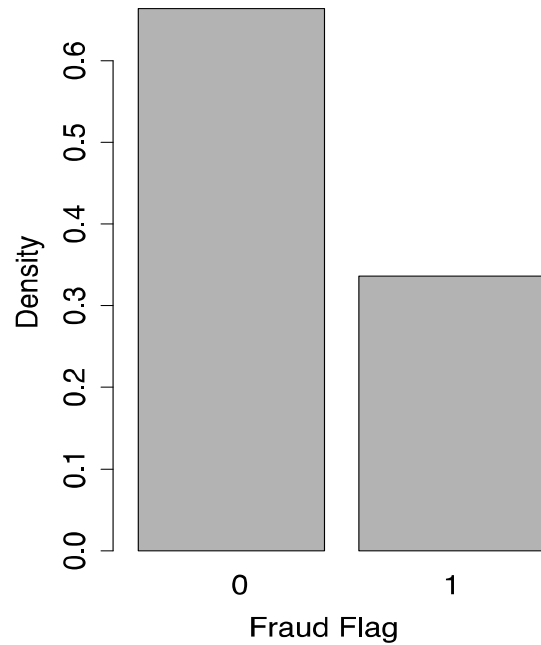


(b) INJURY TYPE

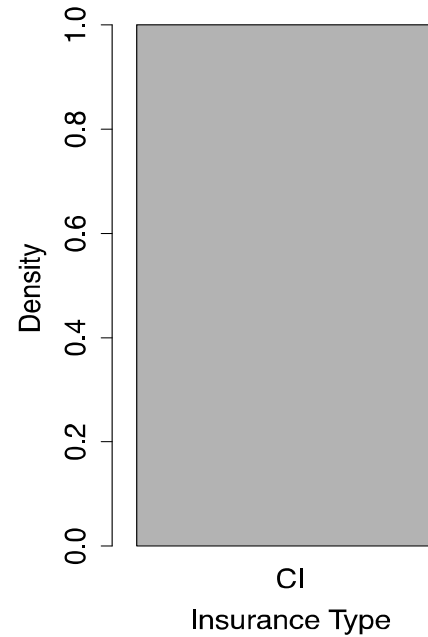


(c) HOSPITAL STAY

Data Quality Report



(a) FRAUD FLAG



(b) INSURANCE
TYPE

Data Preparation

- Data Quality Plan
 - **Data Quality Issues: common steps**
 - Handling Data Quality Issues (errors, missing values, outliers, cardinality)

Identifying Data Quality Issues

Data quality issue: anything *unusual* about the dataset

- Most common data quality issues:
 - missing values
 - irregular cardinality
 - outliers
- Data quality issues can occur due to:
 - **Invalid data:** errors in generating the CSV
 - **Action:** correct the data and regenerate CSV + redo data quality report
 - **Valid data:** due to the problem domain (e.g., changing data gathering process, sensitive data, cost of gathering feature values, changing storage infrastructure)
 - **Action:** record issue in **data quality plan**; record set of actions

Data Quality Plan

- Structure of a **Data Quality Plan:**
Feature, problem, action



Feature	Data Quality Issue	Potential Handling Strategies
—	—	—
—	—	—
—	—	—
—	—	—

A single feature can have multiple problems and actions, so it will be listed several times in the plan

Data Quality Plan

- Starting to draft a Data Quality Plan for Motor Insurance Fraud Prediction

Feature	Data Quality Issue	Potential Handling Strategies
NUM SOFT TISSUE	Missing values (2%)	
CLAIM AMOUNT	Outliers (high)	
AMOUNT RECEIVED	Outliers (high)	

- Note: to notice outliers for numeric features it is easier to look at boxplots, rather than histograms
- Next, we discuss **common handling strategies (actions to prepare the data)**

Data Preparation

- **Data Quality Plan**

- Identifying Data Quality Issues
- **Handling Data Quality Issues (missing values, irregular cardinality, outliers)**

Handling Data Quality Issues

Missing Values

%Miss Column= Percentage of missing values for each feature

We usually try to preserve as much data as possible

- **Approach 1: Imputation:** Replace missing value with reasonable estimate
 - E.g., replace Nan with another value called 'Missing' or replace Nan with the mean or median or mode **where it makes sense**; careful with doing imputation for sensitive personal attributes, eg Age, Gender, etc. **DISCUSS.**
- **Approach 2: Drop rows:** Apply **complete case analysis** if it affects only a few rows (check the rows with missing values and decide to drop or keep)
- **Approach 3: Drop features:** Drop features that have many missing values (more than 50-60%; **before dropping features, first check if the missing values can be inferred or approximated from other columns or data sources**)
- **Approach 4: New features:** Derive a **missing indicator feature** from features with missing values (eg value 0 for not missing and value 1 for missing)

Handling Data Quality Issues

Missing Values

- **Imputation** = replace missing feature values with a plausible estimate
- Most common approach: replace missing feature values with a measure of central tendency of feature (e.g., mean, median or mode) –
- **Note: be careful if the data is multi-modal, as the mean/median doesn't make much sense then**
- Rough guidelines:
 - Not recommended to use imputation on features missing more than 30% of their values
 - We typically drop features missing more than 50-60% of their values (**after investigating the data carefully to see if we can save this feature**)
 - DISCUSS: Does it make sense to impute data in a certain way? E.g., should you replace all missing data for MaritalStatus with Married? If you do, what are the implications?

Handling Data Quality Issues

Irregular cardinality

Card. Column = Number of distinct values present for a feature in the CSV

- **Features with Card=1:** if invalid data, then verify and fix; if valid data, remove the **constant features** as they are not useful for prediction (for a feature to be useful for prediction, its values should have some variability)
- **Low cardinality:** categorical features incorrectly labeled as continuous or vice-versa, e.g., FraudFlag.
 - E.g., Salary = 30, recorded as continuous rather than categorical: 30 stands for 30k-40k category, 40 for 40-50k, etc.
- **High cardinality:** e.g., MaritalStatus has 9 different levels because of annotator inconsistencies: e.g., feature values are “**married, single, divorced, Married, Single, Divorced, M, S, D**”. We need to rename the feature values to a consistent standard, e.g., map values to standard naming “Married, Single, Divorced”, etc. (at data collection this issue can be somewhat resolved by having a fixed choice menu for feature values, rather than a box with free text).

Handling Data Quality Issues

Outliers

- Due to invalid data (e.g., data entry errors)
- Due to valid data (e.g., huge claims)
- **Clamping** = replace all values above/below an upper/lower threshold, thus removing the outliers
- For feature **a**, and thresholds **lower** and **upper**
- Thresholds set based on domain-knowledge or data distribution (e.g., percentiles)

$$a_i = \begin{cases} lower & \text{if } a_i < lower \\ upper & \text{if } a_i > upper \\ a_i & otherwise \end{cases}$$

- Use with care, outliers are often informative for the prediction task. The lower and upper bounds need to be chosen carefully (e.g., How many rows are affected? Look at percentiles).

Example of Clamping

Outliers:

Example thresholds:

$$a_i = \begin{cases} \text{lower} & \text{if } a_i < \text{lower} \\ \text{upper} & \text{if } a_i > \text{upper} \\ a_i & \text{otherwise} \end{cases}$$

a = [-200, -100, 1, 1, 1, 2, 2, 3, 3, 4, 4, 4, 5, 6, 7, 7, 8, 8, 9, 9, 10, 300, 500]

Boxplot values:

max: 500

3rd q: 8.0

2nd q (median): 4

1st q: 2.0

min: -200

iqr: 8 - 2 = 6.0

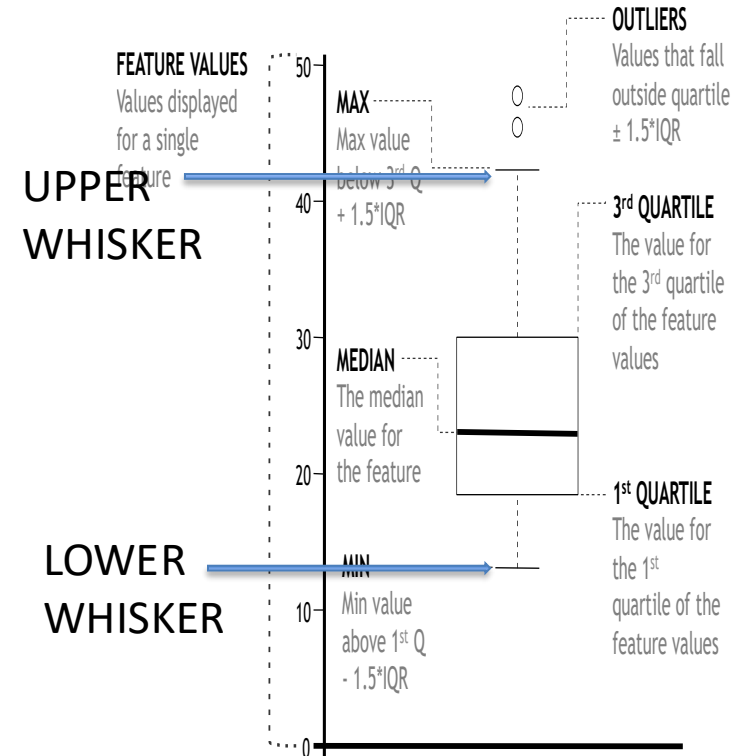
Lower and Upper whiskers are actual values in the data, that are close to the 1st quartile - 1.5 IQR and 3rd quartile + 1.5 IQR

lower: min value in the data that is higher than $2 - 1.5 * 6 = -7.0$ (that is 1 for our data)

upper: max value in the data that is lower than $8 + 1.5 * 6 = 17.0$ (that is 10 for our data)

Clamp **a** to: **a** = [1, 1, 1, 1, 1, 2, 2, 3, 3, 4, 4, 4, 5, 6, 7, 7, 8, 8, 9, 9, 10, 10, 10]

It is always better if we know the meaning of this feature, so we are more informed on how or if to clamp it.



Handling Data Quality Issues

Data Quality Plan for Motor Insurance Fraud (the textbook only presents a starting plan, look at Lab4 for a more detailed plan that lists out each feature).

Feature	Data Quality Issue	Potential Handling Strategies
NUM SOFT TISSUE	Missing values (2%)	Imputation (median: 0.0)
CLAIM AMOUNT	Outliers (high)	Clamp transformation (manual: 0, 80 000)
AMOUNT RECEIVED	Outliers (high)	Clamp transformation (manual: 0, 80 000)

Continuous features:

Issues identified:

Income: Median=0 (50% of values are 0, missing data, **drop Income column**)

ClaimAmount: Min=-99,999 (negative value for claim, **check and fix**)

NumSoftTissue: 2% missing data (imputation to median?, **check and fix**)

TotalClaim: Max=729k (check row/example with this value; **drop row affected**)

Categorical features

Issues identified:

InsuranceType: Card=1 (constant feature, **drop constant column**)

MarritalStatus: 66% missing data (collected together with Income, so 0 Income is actually missing data; **drop column**)

See a more detailed Data Quality Plan in the Lab4 notebook.

Data Understanding: Summary

During the **data understanding/prep** process prepare:

1. Data Quality Report (DQR)

- Get to know the data: tables + plots + notes and discussion (what features to investigate further, actions to take)

2. Data Quality Plan (DQP)

- Actions to prepare & clean data:
 - remove duplicate columns (care needed with duplicate rows: are these errors?, how many rows are duplicates?, how do they affect the analysis?, eg computing the feature stats)
 - decide feature types (continuous or categorical)
 - check for and fix: irregular cardinality, missing values, outliers

Data Understanding: Summary

During the **data understanding/prep** process:

1. Describe the features within the dataset (CSV): central tendencies, variations, and distributions
2. Identify **data quality issues** within the CSV: **missing values**, **irregular cardinality**, and **outliers**; **data integrity checks** (e.g., assert min feature value >0 or check that dates make sense, eg retirement date should be after birth date)
3. Correct data quality issues due to **invalid data**
4. Record data quality issues due to **valid data** in a **data quality plan** along with potential handling strategies
5. **Does enough good quality data exist to continue with a project? If yes, go to further analysis and ML algorithms, if no, go back and get more or better data.**

Lab4

- Continue with the **Motor Insurance dataset** and **Data Quality Report** and **Data Quality Plan** and cleaning the dataset (sample code and sample plan in Lab4 notebook to be discussed at the lab)

References

- **FMLPDA Book: Fundamentals of Machine Learning for Predictive Data Analytics**, John D. Kelleher, Brian Mac Namee, Aoife D'Arcy, MIT Press, 2015 (machinelearningbook.com)